King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Final Exam Term 223

Original exam date: 15 August 2023. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. $\lim_{x\to 0} \frac{\cos x - \sin x - 1}{2x} =$

(a) $-\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) -1

(e) $-\frac{1}{4}$

2. The equation of the tangent line to the graph of the equation $\arctan(xy) = \arcsin(x+y)$ at the point $(0,0)$ is

(a) $y = x$

(b) $y = \frac{1}{2}x$

(c) $y = -x$

(d) $y = 2x$

(e) $y = -2x$

3. The number of vertical asymptotes of the graph of the function $f(x) = \dfrac{x^2-9}{x^3+3x^2-x-3}$ is

(a) 2

(b) 1

(c) 3

(d) 4

(e) 0

4. If $y = \ln\left(\tanh\frac{x}{2}\right)$, then $y'(\ln 2) =$

(a) $\frac{3}{5}$

(b) $\frac{2}{5}$

(c) $\frac{4}{3}$

(d) $\frac{4}{5}$

(e) $\frac{3}{4}$

5. The graph of the function $f(x) = \frac{24}{x^2+12}$ is concave downward on the interval:

(a) $(-\infty, -2)$

(b) $(-2, 2)$

(c) $(-12, -2)$

(d) $(2, \infty)$

(e) $(-12, 12)$

6. $\lim_{x\to\infty} \frac{e^{x^2}}{1-x^3} =$

(a) ∞

(b) e

(c) 0

(d) $-\infty$

(e) 1

7. The absolute maximum of the function $f(x) = \arctan(x^2)$ on the interval $[-2, 1]$ is:

(a) $\frac{\pi}{4}$

(b) $\arctan 2$

(c) 0

(d) $\arctan 3$

(e) $\arctan 4$

8. If $L(x)$ is the tangent line approximation to the graph of $f(x) = 3^x$ at the point $(2, 9)$, then $L(1) =$

(a) $9 + 27 \ln 3$

(b) $8 - 8 \ln 3$

(c) $9 + 9 \ln 3$

(d) 9

(e) $9 - 9 \ln 3$

9. $\int (4t^2 + 3)^2 \, dt =$

(a) $\dfrac{16t^5}{5} + 8t^3 + 9t$

(b) $\dfrac{16t^5}{5} + 6t^3 + 9t + c$

(c) $\dfrac{16t^5}{5} + 8t^3 + 6t + c$

(d) $3t^5 + 8t^3 + 9t$

(e) $\dfrac{16t^5}{5} + 8t^3 + 9t + c$

10. The function $f(x) = \sec \dfrac{\pi x}{4}$ is continuous when

(a) $x \neq 4 + 4n$ where n is an integer

(b) $x \neq 1 + n$ where n is an integer

(c) $x \neq 4 + 2n$ where n is an integer

(d) $x \neq 1 + 4n$ where n is an integer

(e) $x \neq 2 + 4n$ where n is an integer

11. $\lim_{x \to 0^+} 3x^{x/2} =$

(a) 3

(b) 0

(c) 2

(d) 1

(e) $\ln 2$

12. The equation of the tangent line to the graph of the function $f(x) = (\cosh x - \sinh x)^2$ at the point $(0, 1)$ is $y =$

(a) $-x + 1$

(b) $-2x - 1$

(c) $2x + 1$

(d) $-2x + 1$

(e) $2x - 1$

13. The volume of oil in a cylindrical container is increasing at a rate of 150 cubic inches per second. The height of the cylinder is ten times the radius. At what rate (in inches per second) is the height of the oil changing when the oil is 35 inches high? (Hint: The volume of the cylinder is $V = \pi r^2 h$)

(a) $\dfrac{100}{7\pi}$

(b) $\dfrac{200}{9\pi}$

(c) $\dfrac{200}{7\pi}$

(d) $\dfrac{200}{49\pi}$

(e) $\dfrac{100}{49\pi}$

14. The function $f(x) = 12x - x^3$ is increasing on the interval

(a) $(-\infty, -2)$

(b) $(-\infty, 2)$

(c) $(-\infty, \infty)$

(d) $(-2, 2)$

(e) $(2, \infty)$

15. The value of c which satisfies the conclusion of the Mean Value Theorem for the function $f(x) = \sin x$ on $[0, \pi]$ is

(a) $\dfrac{\pi}{5}$

(b) $\dfrac{\pi}{6}$

(c) $\dfrac{\pi}{2}$

(d) $\dfrac{\pi}{4}$

(e) $\dfrac{\pi}{3}$

16. If the function $f(x) = \begin{cases} x^2 + 1, & x \leq A \\ 4x - 3, & x > A \end{cases}$ is differentiable everywhere, then $A =$

(a) 1

(b) -2

(c) 2

(d) 0

(e) -1

17. The number of inflection points for the graph of the function $f(x) = \dfrac{6 - x}{\sqrt{x}}$ is:

(a) 1

(b) 3

(c) 4

(d) 0

(e) 2

18. The slope of the graph of the function $f(t) = \dfrac{\sec t}{t}$ at the point $\left(\pi, \dfrac{-1}{\pi}\right)$ is:

(a) 1

(b) $\dfrac{1}{\pi}$

(c) $\dfrac{-1}{\pi}$

(d) $\dfrac{1}{\pi^2}$

(e) $\dfrac{-1}{\pi^2}$

19. A manufacturer wants to design an open box having a square base and a surface area of 108 square inches. If we want to produce a box with maximum volume, then the volume in cubic inches is

(a) 72

(b) 108

(c) 180

(d) 36

(e) 144

20. The graph of the function $f(x) = 3x + \sin x + 2$

(a) has two horizontal tangent lines

(b) has three horizontal tangent lines

(c) has infinitely many horizontal tangent lines

(d) does not have a horizontal tangent line

(e) has one horizontal tangent line

21. If $4xy + \ln(x^2y) = 7$, then $\dfrac{dy}{dx} =$

(a) $\dfrac{2y - 4xy^2}{4x^2y + x}$

(b) $\dfrac{-4xy^2 + 2y}{4x^2y + x}$

(c) $\dfrac{4xy^2 - 2y}{4x^2y - x}$

(d) $\dfrac{-2xy^2 - y}{4x^2y + x}$

(e) $\dfrac{-4xy^2 - 2y}{4x^2y + x}$

22. If $y = \ln\left(\dfrac{1 + e^x}{1 - e^x}\right)$, then $y' =$

(a) $\dfrac{2e^x}{1 + e^{2x}}$

(b) $\dfrac{2e^{2x}}{1 - e^{2x}}$

(c) $\dfrac{e^{2x}}{1 + e^x}$

(d) $\dfrac{2e^x}{1 - e^{2x}}$

(e) $\dfrac{e^{2x}}{1 - e^x}$

23. Let x and y be two positive numbers such that the sum of the first number cubed and the second number is 500 and the product is a maximum. Then $x + y =$

(a) 420

(b) 380

(c) 400

(d) 340

(e) 360

24. If $y = Ax + B$ is the slant asymptote for the graph of the function $f(x) = \dfrac{x^3 - 3x^2 + 2}{x(x-3)}$, then $A + B =$

(a) -1

(b) 2

(c) 1

(d) 0

(e) -2

25. $\lim_{x \to 2^+} \left(\frac{1}{x^2-4} - \frac{\sqrt{x-1}}{x^2-4} \right) =$

(a) $\dfrac{1}{8}$

(b) $-\dfrac{1}{8}$

(c) 1

(d) $-\dfrac{1}{4}$

(e) $\dfrac{1}{4}$

26. $\lim_{\Delta x \to 0} \frac{(x+\Delta x)^3 - x^3}{\Delta x} =$

(a) $2x^3$

(b) $3x^2$

(c) $-3x^2$

(d) $2x^2$

(e) x^2

27. The measurement of the side of a square floor tile is 10 inches, with a possible error of $\dfrac{1}{32}$ inch. The percent error in computing the area of the square is:

(a) 1%

(b) $\dfrac{8}{3}\%$

(c) $\dfrac{3}{8}\%$

(d) $\dfrac{5}{8}\%$

(e) $\dfrac{8}{5}\%$

28. A ball is thrown upward with an initial velocity of 64 feet per second from an initial height of 80 feet. Assume the acceleration is $a(t) = -32$ feet per second per second. The ball will hit the ground after

(a) 7 seconds

(b) 6 seconds

(c) 4 seconds

(d) 5 seconds

(e) 3 seconds

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\frac{1}{2}\frac{\cos x-1}{x} - \frac{1}{2}\frac{\sin x}{x} \to 0 - \frac{1}{2} = -\frac{1}{2}$.
2	**(c)**	Implicit differentiation at $(0,0)$: $0 = 1 + y'$, so $y' = -1$ and $y = -x$.
3	**(a)**	$f = \frac{x-3}{(x-1)(x+1)}$ after cancelling $x+3$: asymptotes $x = \pm 1$, so 2.
4	**(c)**	$y' = \frac{1}{\sinh x}$, and $\sinh(\ln 2) = \frac{3}{4}$, so $y'(\ln 2) = \frac{4}{3}$.
5	**(b)**	$f'' = \frac{144(x^2-4)}{(x^2+12)^3} < 0$ for $-2 < x < 2$.
6	**(d)**	Exponential beats polynomial; L'Hopital twice gives $-\infty$.
7	**(e)**	Critical point $x = 0$; values $0,\ \frac{\pi}{4},\ \arctan 4$; maximum $\arctan 4$.
8	**(e)**	$L(x) = 9 + 9\ln 3\,(x-2)$, so $L(1) = 9 - 9\ln 3$.
9	**(e)**	Expand: $16t^4 + 24t^2 + 9$, integrate: $\frac{16t^5}{5} + 8t^3 + 9t + c$.
10	**(e)**	$\cos\frac{\pi x}{4} = 0 \iff x = 2 + 4n$; continuous for $x \neq 2 + 4n$.
11	**(a)**	$\ln y = \frac{x}{2}\ln x \to 0$, so $x^{x/2} \to 1$ and the limit is 3.
12	**(d)**	$f = e^{-2x}$, $f'(0) = -2$, so $y = -2x + 1$.
13	**(d)**	$V = \frac{\pi h^3}{100}$, $150 = \frac{3\pi h^2}{100}h'$, so $h' = \frac{200}{49\pi}$ at $h = 35$.
14	**(d)**	$f' = 3(4 - x^2) > 0$ for $-2 < x < 2$.

Q	Ans	Brief explanation
15	**(c)**	Average slope 0, $\cos c = 0$ on $(0, \pi)$, so $c = \pi/2$.
16	**(c)**	Matching slopes $2A = 4$ gives $A = 2$; values agree ($5 = 5$).
17	**(d)**	$f'' = \frac{x+18}{4x^{5/2}} > 0$ on the domain, so no inflection points.
18	**(d)**	$f' = \frac{t\sec t\tan t - \sec t}{t^2}$, $f'(\pi) = \frac{1}{\pi^2}$.
19	**(b)**	$V = \frac{108x - x^3}{4}$, $x = 6$, $h = 3$, $V = 108$.
20	**(d)**	$f' = 3 + \cos x \geq 2 > 0$, so no horizontal tangent.
21	**(e)**	$y'(4x + \frac{1}{y}) = -4y - \frac{2}{x}$, so $y' = \frac{-4xy^2 - 2y}{4x^2y + x}$.
22	**(d)**	$y = \ln(1 + e^x) - \ln(1 - e^x)$, $y' = \frac{2e^x}{1 - e^{2x}}$.
23	**(b)**	$P = 500x - x^4$, $x = 5$, $y = 375$, so $x + y = 380$.
24	**(c)**	$f = x + \frac{2}{x^2 - 3x}$, slant asymptote $y = x$, $A + B = 1$.
25	**(b)**	$\frac{1 - \sqrt{x-1}}{x^2 - 4}$, L'Hopital: $\frac{-1/(2\sqrt{x-1})}{2x} \to -\frac{1}{8}$.
26	**(b)**	Expand: quotient $= 3x^2 + 3x\Delta x + (\Delta x)^2 \to 3x^2$.
27	**(d)**	$\frac{dA}{A} = \frac{2dx}{x} = \frac{1}{160}$, i.e. $\frac{5}{8}\%$.
28	**(d)**	$s = -16t^2 + 64t + 80 = 0 \Rightarrow t = 5$ seconds.